

MINOR PROJECT PYTHON-1:

SUPERMARKET BILL GENERATION USING PYTHON

Introduction

This program is a **basic billing system** written in Python. It allows a shopkeeper to enter a customer's name, display a list of available items with prices, let the customer choose items and quantities, and finally generate a bill with GST (tax) included.

Code Explanation

1. Importing Date and Time

```
python
```

```
from datetime import datetime
```

- This line imports the datetime module so the program can print the current date and time on the bill.

2. Taking Customer Name

```
python
```

```
name = input("Enter Customer Name: ")
```

- The program asks the user to type the customer's name.
- This name will be shown on the final bill.

3. Displaying Item List

```
python
```

```
lists = '''
```

```
Rice  Rs 20/kg
```

```
Sugar Rs 30/kg
```

```
Salt  Rs 20/kg
```

```
Oil   Rs 80/liter
```

```
Paneer Rs 110/kg
```

```
Maggi Rs 50/kg
```

```
Boost Rs 90/kg
```

```
Colgate Rs 85/kg
```

```
'''
```

```
print(lists)
```

- A list of items with their prices is stored in a string called lists.
- The program prints this list so the customer can see what is available.

4. Initializing Variables

```
python
```

```
price = 0
```

```
pricelist = []
```

```
totalprice = 0
```

```
finalamount = 0
```

```
quantitylist = []
```

```
itemlist = []
```

- These variables keep track of:
 - price: cost of one item purchase
 - pricelist: list of prices for each chosen item
 - totalprice: sum of all items before tax
 - finalamount: total after adding GST
 - quantitylist: how many units of each item were bought
 - itemlist: names of items purchased

5. Defining Item Prices

```
python
```

```
items = {
```

```
    'rice': 20,
```

```
    'sugar': 30,
```

```
    'salt': 20,
```

```
    'oil': 80,
```

```
    'paneer': 110,
```

```
    'maggi': 50,
```

```
    'boost': 90,
```

```
    'colgate': 85
```

```
}
```

- A dictionary called items stores the price of each product.
- Example: "rice": 20 means rice costs Rs 20 per kg.

6. Showing Items Again (Optional)

```
python
```

```
option = int(input("For list of items press 1: "))
```

```
if option == 1:
```

```
    print(lists)
```

- If the user presses 1, the program prints the item list again.

7. Buying Items

python

```
for i in range(len(items)):
```

```
    inp1 = int(input("If you want to buy press 1 or 2 for exit: "))
```

```
    if inp1 == 2:
```

```
        break
```

```
    if inp1 == 1:
```

```
        item = input("Enter Your Item: ").lower()
```

```
        quantity = int(input("Enter your quantity: "))
```

```
        if item in items.keys():
```

```
            price = quantity * items[item]
```

```
            pricelist.append(price)
```

```
            totalprice += price
```

```
            itemlist.append(item)
```

```
            quantitylist.append(quantity)
```

```
        else:
```

```
            print("Sorry, the item you entered is unavailable.")
```

```
    else:
```

```
        print("You entered wrong number")
```

- The program loops through items and asks if the customer wants to buy.
- If they press 1, they enter the item name and quantity.
- The program calculates the price (quantity × rate) and adds it to the bill.
- If the item is not in the list, it shows an error message.
- Pressing 2 exits the buying process.

8. Billing Section

python

```
inp = input("Can I move to billing? Yes or No: ")
```

```

if inp.lower() == 'yes':
    if totalprice != 0:
        gst = (totalprice * 5) / 100
        finalamount = gst + totalprice
        print("\n-----BILLING-----")
        print("Customer Name:", name)
        print("Date:", datetime.now())
        print("-----")
        for i in range(len(itemlist)):
            print(f'{itemlist[i]} x {quantitylist[i]} = Rs {pricelist[i]}')
        print("-----")
        print("Total Price:", totalprice)
        print("GST (5%):", gst)
        print("Final Amount:", finalamount)

```

else:

- ```

print("No items purchased.")

```
- After shopping, the program asks if it should move to billing.
  - If the answer is “yes” and items were purchased:
    - It calculates **GST = 5% of total price**.
    - Adds GST to get the **final amount**.
    - Prints a neat bill with:
      - Customer name
      - Date and time
      - Item details (name × quantity = price)
      - Total price, GST, and final amount.
  - If no items were bought, it prints “No items purchased.”

### **Conclusion**

This program is a **simple billing system** that:

- Shows available items and prices.
- Lets the customer choose items and quantities.
- Calculates the total cost with GST.

- Prints a formatted bill with all details.

It's a good starting point for building a small shop billing application.